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Deep Learning-Based Forecasting of Groundwater Level Trends in India: Implications for Crop Production and Drinking Water Supply

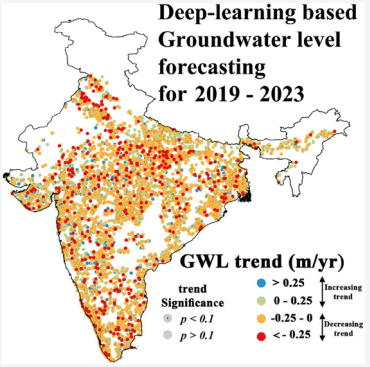
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ABSTRACT: Despite numerous studies in recent times, there is no consensus on the primary drivers for groundwater storage (GWS) changes over India. Thus, predicting future groundwater level trends seems remote. In this context, using Gravity Recovery and Climate Experiment (GRACE)-derived GWS, WaterGap model-based groundwater recharge (GWR), and groundwater withdrawal (GWW), we show that GWW exhibits a stronger dominance than GWR on GWS change over India. Furthermore, we developed feed-forward neural network (FNN), recurrent neural network (RNN), and deep learning-based long short-term memory network (LSTM) models using multidepth in situ observations from a dense network of monitoring wells (n = 5367, 1996−2018), to simulate and forecast groundwater levels (GWL) in India. The result demonstrates the better performance of LSTM (>84% of observation wells showing $r > 0.6$, $RMSE_n < 0.7$) across India, outperforming both FNN and RNN in the testing period of 5 years (2014−2018). Our estimates also reveal that besides the prevailing long-term (1996−2018) statistically significant ($p < 0.1$) declining GWL trends in northwest India and the Ganges river basin, higher declining trends will potentially be observed in parts of north-central and south India in the forecasting period of 5 years (2019−2023). We envisage that the forecasting approach presented in the study can contribute toward an improved urban−rural drinking water supply and sustainable crop production for 1.3 billion people in India.



KEYWORDS: *Groundwater quantity, Relative driver importance on groundwater storage, LSTM-based forecasting, water−food−energy nexus, India*

1. INTRODUCTION

India is the largest consumer of groundwater globally, accounting for ∼25% of global groundwater withdrawals.[1,2] Over 63% of the irrigated land (>30% of the total area) in India is groundwater reliant,[3,4] contributing ∼9% of India's GDP.[5] The extensive agricultural development linked to groundwater-fed irrigation for yearly multiple crops (i.e., rabi and kharif) has caused significant groundwater storage depletion, notably in the Indus−Ganges−Brahmaputra river basins. Despite considerable higher annual rainfall (∼1200 mm/yr), the majority (∼75%) of rainfall occurs during the monsoon season (June−September),[6,7] leading to monsoon-dependent groundwater recharge.[8] Hence, India's groundwater storage variability is governed by the dynamic equilibrium between the components of the nature−human composite system. Thus, forecasting the groundwater quantity is challenging due to the high complexity of the system. Despite of several studies, there is no consensus on the influence of the major drivers in groundwater storage and/or recharge change over India. Few studies[9,10] and their attributions emphasized the strong linkage between precipitation and groundwater storage change in India. Other studies[1,5,11,12] report that intensive groundwater withdrawal strongly controls groundwater storage, and even if increased precipitation is observed, groundwater storage started diminishing in parts of India.[8] Furthermore, studies on the Indus−Ganges−Brahmaputra basin[13] and the Bengal basin[14,15] identified abstraction as the significant trigger enhancing primary or secondary recharge. Among other aspects, groundwater storage is found to be directly linked to streamflow reduction,[16] influenced by policy change,[1] replenished by irrigation return flow,[5] and also controlled by the pore-elastic effect.[17] In this context, developing a coherent understanding of groundwater systems and forecasting the future availability of groundwater storage is essential for sustainable planning, preparedness due to hydro-climatic alterations,[18] and avoiding a socio-economic impact[19] in India, where the population is rapidly expanding.

In recent times, considering the availability of large quantities of high-quality spatiotemporal data, data-driven

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approaches have been used to solve complex hydrologic problems and the prediction of resources. The AI-based methods augment the existing conceptual, numerical, and physically based models due to their ability to discover physically meaningful relationships within hydrologic data and automatic extraction of spatiotemporal features.[20] The complex hydrologic relationships can be modeled through advanced data-driven approaches since the processes are embedded in the time series of the observations.[21] The GWL variations in the wells provide a more direct measure of the impact of groundwater development and information on aquifer dynamics. Thus, over the years, several AI-based modeling techniques (feed-forward neural network (FNN), recurrent neural network (RNN), support vector machine (SVM), convolutional neural network (CNN), long short-term memory network (LSTM), and others) have been applied by the researchers for groundwater resource prediction (and simulations) across the world.[22−32]

Bowes et al. (2019)[33] explored the ability of RNN and LSTM to model GWL in the city of Norfolk, VA, and identified LSTM as outperforming RNN. Using daily GWL data from 2003 to 2008 of five observation wells in South Korea, Yoon et al. (2016)[34] developed a weighted error function approach to increase the performance of ANN and SVM-based recursive prediction models for the long-term GWL prediction in response to precipitation. The comparison between ANN and SVM suggests that SVM performs better than ANN. Daliakopoulos et al. (2005)[35] tested seven ANN architectures to simulate the decreasing GWL trend up to 18 months ahead in Messara Valley in Greece and reported that FNN trained with the Levenberg−Marquardt performs better than other FNN and RNN architectures. Yoon et al. (2011)[36] reported that SVM has better accuracy than ANN in GWL prediction in the coastal aquifer of South Korea and identified the tide level as the most influencing input. Furthermore, in the U.S.A.'s agriculture aquifers, Sahoo et al. (2017)[37] reported that, using preprocessed GWL data and others, hybrid ANN is superior to hybrid MLR in predicting GWL changes. In another study with preprocessed meteorological data, Jeong et al. (2019)[38] noted that LSTM and a nonlinear autoregressive network with exogenous inputs (NARX) offer a good performance in predicting GWL in two in situ wells. Zhang et al. (2018)[28] applied LSTM to identify the stressed condition in an aquifer of Hetao Irrigation District, China, through a comparison between the observed and predicted GWL. A recent study by Yin et al. (2021)[39] developed ANN, SVM, and response surface regression models and reported the superiority of an ensemble modeling framework with a Bayesian model averaging approach for groundwater storage change prediction in the San Joaquin River Basin. A study by Rahman et al. (2020)[40] developed Random Forests and eXtreme Gradient Boosting and combined with wavelet transformation to predict 1−3 months ahead GWL forecasts and presented good accuracy with the observed values in seven observation wells in Kumamoto City, Japan. The novel methodologies presented in most previous literature have significant implications on regional to local groundwater resource management. However,most of these studies across the world are based on a limited geographical area and relatively small temporal length. Thus, the large spatial and long-term variability of groundwater resource prediction is not investigated yet with high-resolution, field-scale observations using AI-based models.

A few previous studies adopted AI-based techniques in parts of India for groundwater resource prediction. In a study based on Bengaluru, India, Yadav et al. (2020)[41] predicted GWL using ANN and SVM on preprocessed GWL, climatic, and population data and revealed the effects of population growth on GWL changes. Using the linear regression model (LRM), SVM, and ANN, a recent study predicted GWL in a few (n = 5) monitoring wells in India with GRACE-derived TWSA and climatic variables. They reported better performance of SVM using GRACE-derived TWSA and climatic variables together as input.[42] Sun et al. (2019)[43] compared GRACE-derived TWSA and CNN simulated land surface model based TWSA in India. However, they were using either a short temporal and little spatial extent[41,42] or did not incorporate high-resolution field-scale observations.[43] Due to the heterogeneity across India, to develop a country-scale GWL forecasting model, it is essential to examine the model performance spatially and aquifer depth.

Most previous literature (in India and the rest of the world) tested the ability of the AI-based models with a comparison between observed and predicted (or simulated) values of groundwater quantity (i.e., GWL and/or groundwater storage). The forecasting of the groundwater quantity into the future (i.e., unseen data into the future) has been relatively less attempted. A recent study by Wöhling et al. (2020)[44] proposed 30 days operational forecast of GWL using the eigenmodel approach. Abuamra et al. (2020)[45] used few forecasting algorithms such as an artificial neural network (ANN) and autoregressive integrated moving average (ARIMA) to forecast GWL in two observation wells in the Gaza Strip. Mackay et al. (2015)[46] used seasonal weather forecast system outputs to forecast GWL for up to three months. Furthermore, using the forecasted values of the precipitation and sea level as input data, Bowes et al. (2019)[33] carried out a real-time GWL forecasting scenario and developed LSTM models to construct hourly forecasts of GWL. Mohanty et al. (2015)[47] used weekly precipitation, pan evaporation, surface drain water level, river stage, GWL, and pumping rates to predict GWL at 2 to 4 weeks ahead at 18 sites and concluded shorter lead time results are better. Using a similar input, Mohanty et al. (2013)[48] developed ANN and numerical models to predict GWL and find higher accuracy for ANN models. They also perform predictive numerical simulations to simulate GWL in the long run under prevailing pumping conditions. Under this scenario, they explored the impact of pumping on GWL between 2007 and 2020, keeping all the prevailing conditions constant. These studies are mainly local studies and/or have not pursued a longer ahead time forecasting, which is required to quantify large-scale dynamical changes in future groundwater availability detection and attribution. The medium- to long-term forecasting[45] provides (i) informed strategies to avoid or reduce an adverse impact due to groundwater depletion,[49] which is severe in most of India;[1] (ii) identifying the dynamics and underlying drivers leading to depletion of groundwater storage and taking countermeasures; (iii) maintaining a balance between supply and demand in the agricultural (∼60% irrigated agriculture in India depends on groundwater), domestic (85% of drinking water supply in India),[50] and industrial sectors; (iv) analyzing the advantages and expenses of water conservation[51] for sustaining the vast population of India.

Despite the critical importance of a credible understanding of the groundwater system, the relative role of the significant

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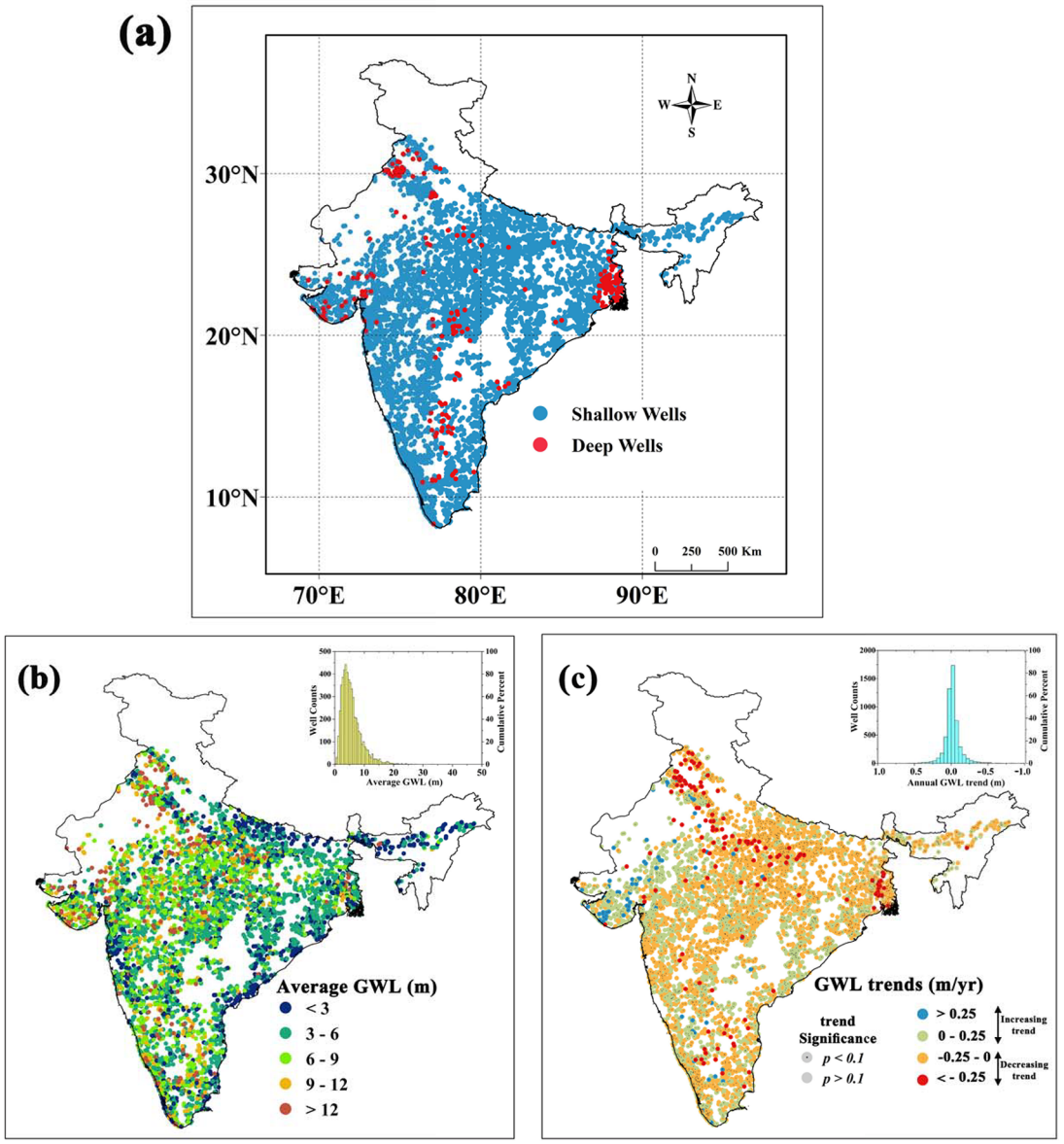


Figure 1. (a) Map of the study area that shows the locations of the 5367 groundwater observation wells (shallow observation wells, n = 5210; deep observation wells, n = 247). (b) Average groundwater level (m bgl) map between 1996 and 2018. (c) Trends in the in situ groundwater level determined by Mann−Kendall trend tests for slope significance and Sen's slope estimator for slope magnitude from 1996 to 2018.

natural to human-induced drivers and estimation of future groundwater trends in India remains largely unrecognized. So far, all published studies have done depth-integrated analyses of the groundwater response to extraneous factors, however, doing so the signals of the groundwater response because of the multidepth variability and the consequent groundwater availability are not delineated. Thus, the renewable groundwater (through dynamic groundwater recharge, shallow groundwater) vs nonrenewable (i.e., deep) groundwater is not taken into account for calculation, and accordingly the predictions do not provide a complete representation of the system. This leads to a lack of an extremely important required input for the prediction of future groundwater resource estimation. In this study, we have tried to disentangle the depth-dependent signal to identify the true groundwater stress and accordingly predict the potential future availability of groundwater for food production and drinking water in India, under changing climate and human scenarios. Thus, the novelty of the study is based on some of the research gaps not explored in previous studies: first, we used depth-dependent GWL (i.e., using the screen depths of the observation wells, shallow vs deep groundwater) to estimate the GWL trend with Theil−Sen method and Mann−Kendell test at different depths of the aquifer for the first time. Second, using multiple regression-based dominance analysis on various data sources (gravity recovery and climate experiment (GRACE) based GWS, WaterGap global hydrology model-based groundwater recharge (GWR), and groundwater withdrawal (GWW)), we explored the relative influence of GWW and GWR on GWS across India. Third, we compared the performance of deep learning (DL)-based LSTM, FNN, and RNN to simulate multidepth GWL for an extensive network of monitoring wells across India and linked them to the natural to human-induced drivers. Fourth, using the best performing AI based method, we forecasted depth-dependent GWL trends (i.e., shallow vs deep groundwater) for future times and compared it with the

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past GWL trends. Lastly, based on the findings and further analyses, we discussed their implications for future crop production and drinking water supply for 1.3 billion people in India.

The primary objective of our work is to forecast future trends of GWL changes at different depths of the aquifer across India. To do that, we

(a) estimated the long-term trends of in situ groundwater levels of a large monitoring network (n = 5367) and at different depths of the aquifer (shallow vs deep groundwater) and quantified the influence of the primary drivers on GWS at present times
(b) developed and determined the most suitable model among FNN, RNN, and DL-LSTM that can learn the patterns of spatial-temporal variations associated with the present in situ GWL changes to forecast GWLs into the future
(c) compared the prevailing and forecasted GWL trends (shallow vs deep groundwater) across India and their significance for future crop production and drinking water supply.

2. MATERIALS AND METHODS

2.1. Study Area. India is the largest country in South Asia, with regard to global land area (~2.3%) and global population (~17.8%).[3] India has an extremely heterogeneous hydrogeologic setting (Figure S1a), precipitation distribution (Figure S1b), groundwater abstraction (Figure 3a), recharge pattern (Figure 3b), climate, and vegetation cover.[5,6,52] The mean annual precipitation patterns exhibit a wide variation from extreme humid regions (annual precipitation > 2500 mm/yr) in the east and southwestern coast to the arid areas (annual precipitation < 700 mm/yr) in the west, which contributes to the spatially diverse groundwater recharge.[5] The major river basins of India are shown in Figure S1d.

2.2. Data Selection. Groundwater level (GWL) data from 1996−2018 for more than 26 000 observation wells have been used in this study. The screen depth of the observation wells ranges from 3 m to 600 m. The well type varies from dug wells and bore wells to tube wells. The GWL data is in a quarterly structure, i.e., measurements four times a year for August (monsoon), November (early postmonsoon), January (late postmonsoon), and May (premonsoon). Terrestrial water storage (TWS) anomalies from January 2003 to December 2014 (133 months) have been obtained from the Jet Propulsion Laboratory, National Aeronautics and Space Administration (NASA).[53,54] To retrieve GWS anomalies, other significant components of TWS, i.e., surface water equivalents and soil moisture equivalents, assimilated from Global Land Data Assimilation System (GLDAS)[55] have been removed from TWS anomalies. Lastly, in order to remove the associated bias, a combination of three models (variable infiltration capacity (VIC), NOAH, and community land model (CLM2)) have been used.[11,52]

The global hydrological model WaterGAP (water−global assessment and prognosis) provides both water resources and human water use on all global land areas except Antarctica. We used the global scale groundwater stress indicator at a spatial resolution of 0.5° × 0.5° (~55 × 55 km in the equation) calculated by the WaterGAP 2.2b model.[56−59] The stress indicators were calculated for 1981−2010 from a homogenized version of the concatenated WFDERA-Interim data sets (WFDEI) and WATCH forcing data ERA-40 (WFD). Here, we used monthly diffuse groundwater recharge and water withdrawals from groundwater (transWU model run), i.e., transient water use in all sectors. A more detailed description can be found in the Supporting Information Text S1.

2.3. Selection Criteria, Data Preprocessing, and Classification. The acquired GWL data consists of a broad range of discontinuity, temporal frequency, and time intervals. Thus, to maintain the continuity and quality within the data set, the quarterly GWL data set has been screened, resulting in 5367 observation wells with ≥3/4 observations for all 23 years (Figure 1a). The raw GWL data has been culled with Tukey's method to discard (~3.65% of total data) outliers.[60] To fill the missing data in the GWL data set (~8.29% of total data), multiple imputation[61−63] is used. The GWL observation wells used in this study were classified based on their screen depth: SH (shallow observation wells; screen depth < 30 m) and DP (deep observation wells; screen depth > 30 m). These categorizations of the observation wells were based on the screen depths, adapted to the prevailing depth of the irrigation wells[64] in India. The well count statistics for final observation wells are provided in Figure S2. Based on the census report,[65] we further categorized "dug wells" as SHs, and "tube wells" and "bore wells" are categorized as DPs.[66]

2.4. Groundwater Level Trend. Yearly groundwater level trends for each observation well were calculated over 1996−2018 using the Theil−Sen method.[67] The statistical significance of the GWL trends was estimated with the Mann−Kendell test.[68] GWL records associated with $p < 0.1$ are considered in the rejection of the null hypothesis, i.e., no significant trend ($\tau = 0$). The Theil−Sen trend estimator is used in combination with the Mann−Kendall test because of its robustness to outliers and nonlinearity.

2.5. Dominance Analysis (DA). Dominance analysis (DA)[69,70] is one of the most employed metrics of relative importance estimation due to its robust approach across all likely subset regression models. The DA method partitions the overall multiple regression's coefficient of determination (R^2) into "shares", which are attributable to each of the predictors,[71] and identifies the predictors that are more or less important or dominant than others. In this study, gridded monthly diffuse groundwater recharge (GWR) and groundwater withdrawals (GWW) are considered independent variables to identify the spatially dominant relationships with GRACE GWS. For a total p submodels, the conditional dominance of the variables for $p - 1$ submodels is performed.[72] A detailed description of DA can be found in Budescu and Azen (2004).[73]

2.6. Long Short-Term Memory Network (LSTM) Model, Development, and Performance. LSTM is a kind of RNN that has the ability to learn long-term dependencies directly from time-series data. The LSTM architecture is designed to overcome issues of exploding and vanishing gradients, which were considered as a serious demerit of the traditional RNN. Furthermore, LSTM maintains stable states for a more extended time period while keeping temporal dependencies.[74] In this study, the sequence to sequence LSTM architecture has been used. A description of the architecture and process of an LSTM network can be found in Supporting Information Text S2.

Here, in this study, we developed a two-layer LSTM model with fully connected hidden layers. The number of hidden layers is set at 300. The forecast time step is set for 1, 2, 3, ..., 40. The maximum epoch is set for 200. The batch size,

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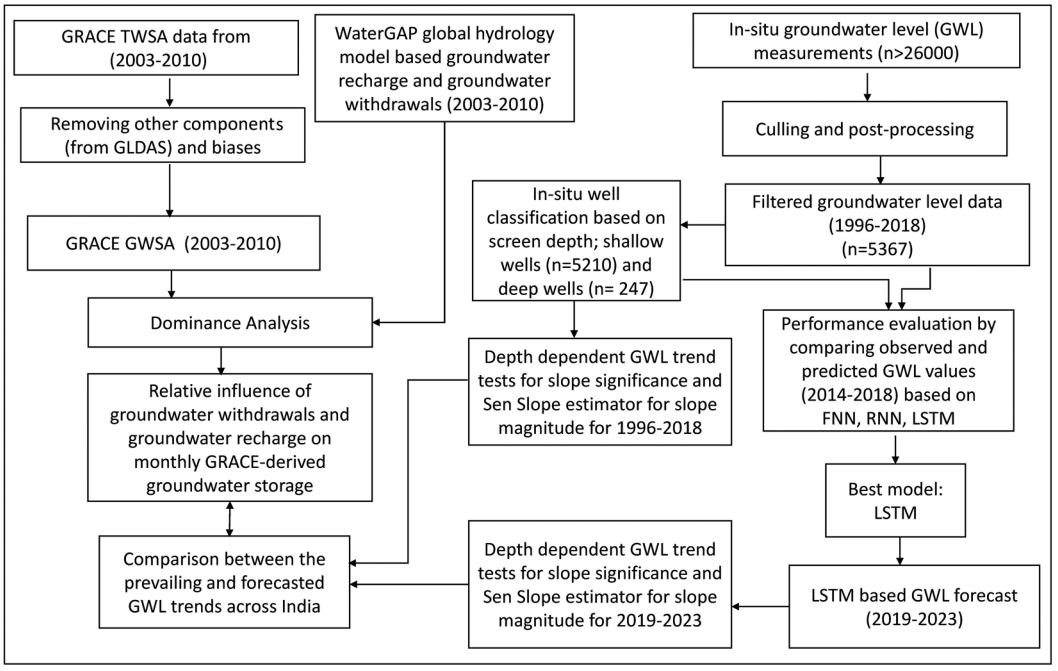


Figure 2. Flowchart showing the methodology followed in this study.

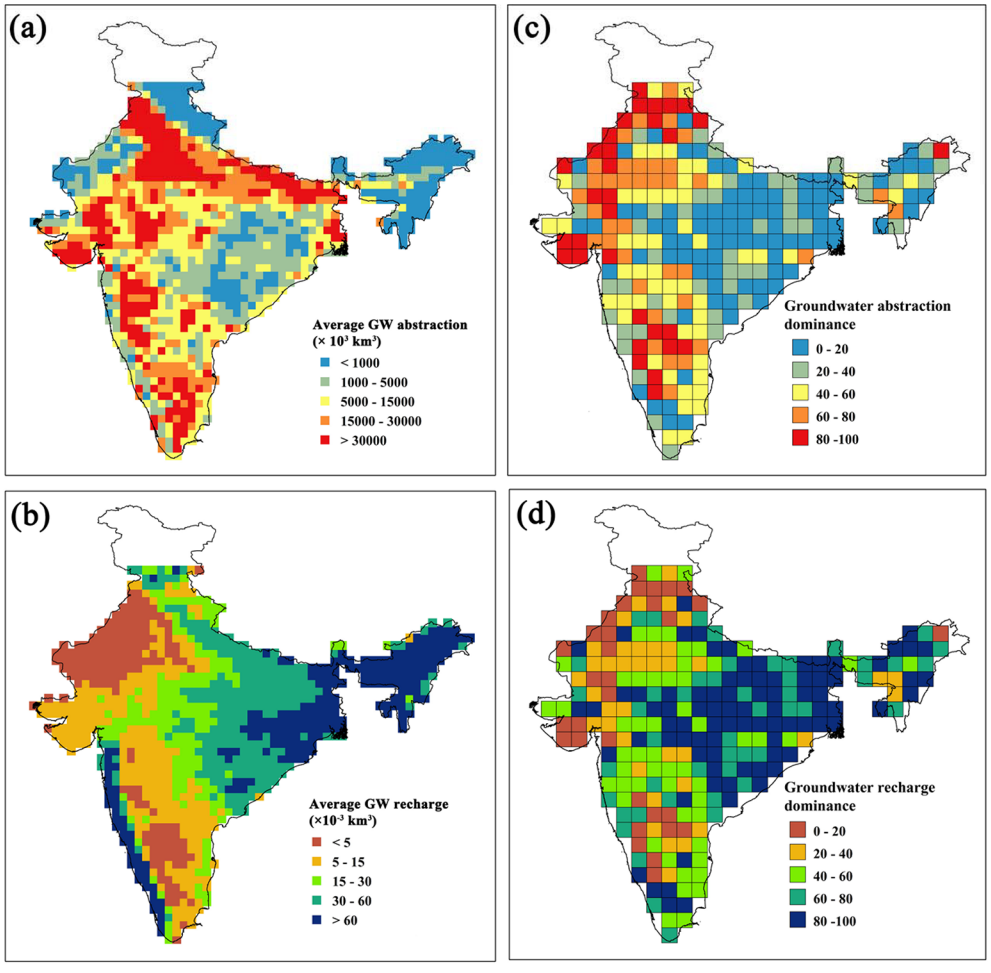


Figure 3. WaterGap global hydrologic model-based (a) average groundwater withdrawals ($\times 10^3$ km^3) and (b) average groundwater recharge ($\times 10^3$ km^{-3}) between 1981 and 2010; relative influence (in %) of (c) groundwater withdrawals and (d) groundwater recharge on monthly GRACE-derived groundwater storage as determined by dominance analysis.

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learning rate, and dropout rate are 50, 0.001, and 0.1, respectively. Adaptive moment estimation (Adam)[28,75] was employed for model optimization, and mean squared error (MSE) was adapted as the loss function. Here, we develop the LSTM model to simulate and forecast GWLs for future times, using only GWL as the input (Table S1). The total GWL data set of 23 years (23 × 4 = 92 seasons, 1996−2018) is split into two parts: the training set (1996−2013, 18 years) and the testing data set (2014−2018, 5 years) and finally predicted GWL for the next 5 years (2019−2023). This model architecture and hyperparameters were adapted in an ad hoc manner based on a trial and error procedure, resulting in the lowest root mean squared error (RMSE)[33] (Supporting Information Text S3). The training and testing of the LSTM models have been executed for each of the 5367 observation wells, and the screen depths of the observation wells were not included in the models. The performance of the LSTM models is evaluated using a correlation coefficient (r) and normalized root-mean-square error ($RMSE_n$) (Supporting Information Text S4). The LSTM model was developed in Matlab 2017b.

The variations in the GWL data set may cause adverse effects on the model learning ability. Thus, the GWL data needs to be standardized to remove the scale dependency and provide stable convergence in model development.[29] The GWL data are standardized (x_{stand}) using mean (x_μ) and standard deviation (x_σ):

$$x_{stand} = \frac{x - x_\mu}{x_\sigma} \quad (1)$$

A detailed flowchart of the methodology adapted in this study is shown in Figure 2.

2.7. Assumptions, Limitations, and Uncertainty. We were limited to using the maximum possible temporal data range from different sources for the analysis that requires multiple data sets (i.e., dominance analysis). Groundwater recharge data taken from the WaterGAP model consists of only diffuse recharge, and recharge from other sources (e.g., irrigation return flow, surface water bodies, etc.) are not included here (Supporting Information text S1). The northernmost regions of India are not included in the study due to the unavailability of data. The spatial data coverage of DP wells is deficient and clustered. In order to fill in the missing data, the multiple imputation method has been used. We assumed the wells (>87%) to be unconfined, as reported.[76]

3. RESULTS AND DISCUSSION

3.1. Depth-Dependent Groundwater Level and Trend Variability. Understanding the present changes in groundwater resources is a prerequisite to estimating the changes in future times. Hence, in this study, first, we assess the present-day groundwater resources using the Sen's slope estimator for trend magnitude and the Mann−Kendall trend test for identification of a monotonic trend significance and mean groundwater level depths at different depths of the aquifer for the first time across India. The mean GWL (1996- 2018) depth for both shallow (SH) and deep (DP) wells exhibit significant variability across the country (Figure 1b). More than 65% of the SH wells show GWLs less than 6 m below ground level (bgl). However, more than 64% of the DP observation wells show GWL deeper than 6 m and more than 30% deeper than 12 m bgl. The deepest GWLs are mostly found in northwest India (Indus basin), eastern India (Bengal basin area of the Ganges basin), and some parts in southern India (parts in Pennar and Cauvery basin) (Figure 1b). Furthermore, the long-term (1996−2018) GWL trend across India shows that, among 5120 SH observation wells, 44% demonstrates statistically significant (p < 0.1) Sen's slope estimates (i.e., trend magnitude) (Figure 1c). However, about 61% of the analyzed DP wells are found to have a statistically significant (p < 0.1) GWL trend. We noted that ∼56% of the analyzed observation wells show near stable (−0.05−0.05 m/yr) GWL trends. However, among those near stable wells, the majority (∼67%) show declining trends. Noticeable GWL depletion trends are found in several parts of the country, primarily in northwest India and parts of the Ganges basin in eastern India, where observation wells have unprecedented GWL decline rates as high as >1 m/yr. Extreme declining trends in northwest India and parts of Ganges basin including Bengal basin area (Figure 1c) are well recognized and subjected to high GWW (Figure 3a) due to crop water requirements[13,77] and drought.[78] However, groundwater rising trends in parts of western India and southern India are attributed to water use management policy change.[1]

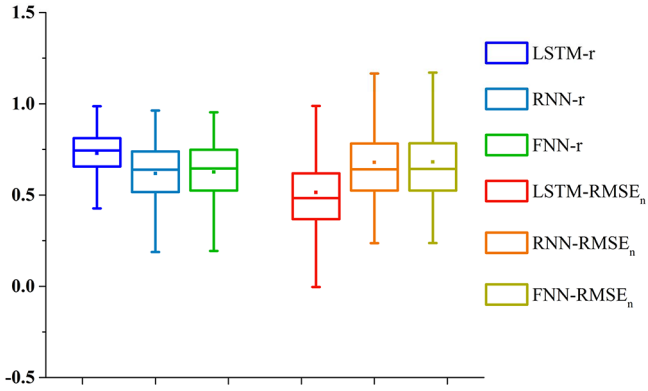


Figure 4. Box plots showing the range country-scale correlation coefficient (r) and normalized root mean square errors ($RMSE_n$) between actual and predicted groundwater levels from individual well-scale analysis for the testing period (2014−2018) in the case of LSTM, RNN, and FNN, respectively.

3.2. Relative Importance of Groundwater Withdrawals and Groundwater Recharge on Groundwater Storage. We estimated the relative importance of groundwater withdrawals and groundwater recharge on groundwater storage using dominance analysis. The dominance analysis findings suggest (Figure 3c) strong dominance of GWW in northwest India (i.e., Indus basin), western India, and parts of the Krishna, Cauvery basin in south India. However, GWR's dominance is higher, mostly in eastern India, including the Brahmaputra basin, extending to the mid-Ganges basin and Mahanadi basin (Figure 3d). Despite high precipitation (Figure S1b) and recharge rates (Figure 3b), the Brahmaputra basin also exhibits GWS depletion linked to high consumptive groundwater abstraction.[8] In India, the impact of the significant reduction in GWS is reflected in 22% of the administrative blocks, which are either overexploited (abstraction > recharge) or in critical (abstraction ∼90% of recharge) condition.[2] Moreover, based on the findings of dominance analysis, the strong influence of GWR (Figure 3d) in eastern India could be explained by the high precipitation (Figure S1b), leading to high direct groundwater recharge (Figure 3b). Precipitation contributing to direct GWR in India[79] results

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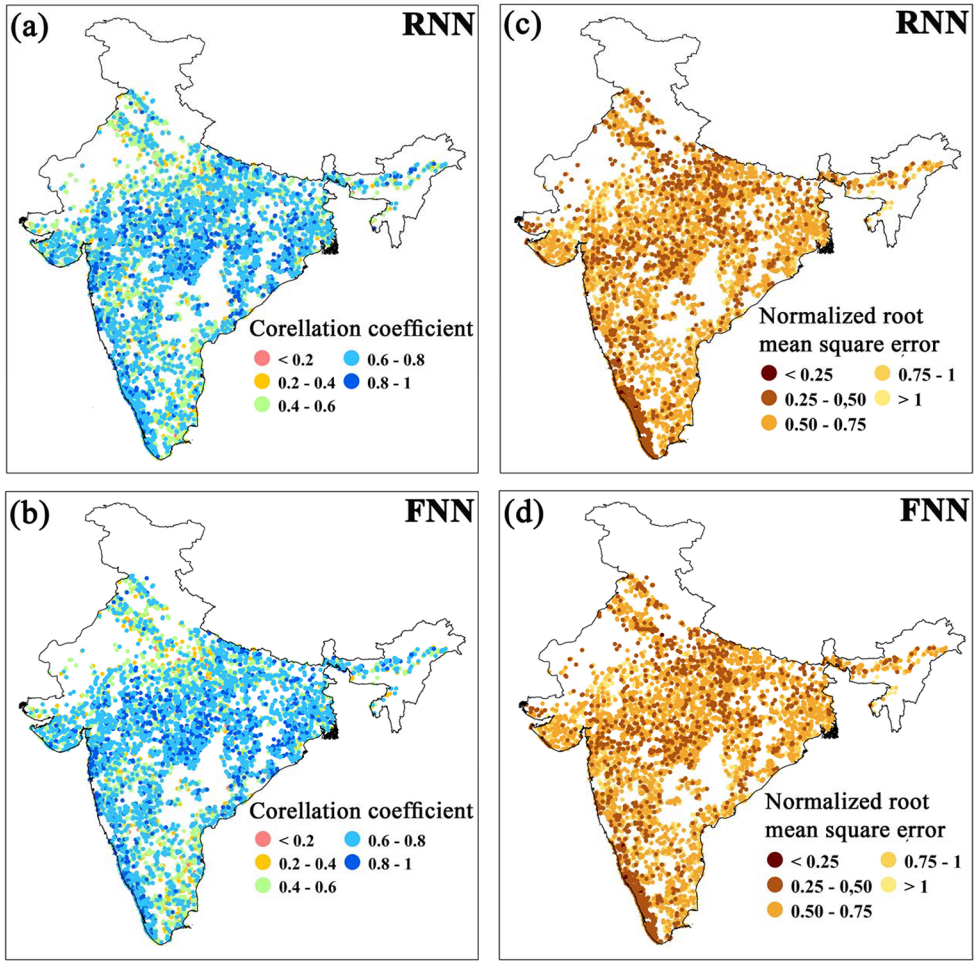


Figure 5. Maps showing the correlation coefficient (a), (b); and Normalized Root mean square error (c), (d) between actual and predicted groundwater levels from individual well-scale analysis for the testing period (2014−2018) in case of RNN and FNN respectively. Levenberg−Marquardt (LM) algorithm was in building both FNN and RNN. The FNN and RNN models have been trained using data for 72 seasons (18 years, 1996−2013, training period) to predict GWLs for the next 20 seasons (5 years, 2014−2018, test period). A detailed description of different types of AI methods in GWL modeling can be found in Rajaee et al. (2019).[89]

from moderate to heavy intensity short episodic precipitation rather than low intensity, distributed precipitation.[80] Moreover, high recharge and precipitation may lead to the re-equilibration of groundwater, which is reflected in the majority of the SH wells exhibiting a shallower GWL (0−6 m) (Figure 1b) in recharge dominated regions except in the Bengal Basin, which is subjected to abundant GWW. GWR's lower dominance in northwest India, western India, and south-central India is an artifact of the smaller amount of recharge (and precipitation), obtained from the WaterGap global hydrologic model. On the contrary, these regions are spatially correlated with higher GWW (Figure 3a). Thus, they are GWW dominated (Figure 3c).

The ascendency of GWW and resulting GWS depletion in India is well documented.[1,13,81] However, we needed to restrict the study up to 2010 for WaterGAP data availability. GRACE had continuous data until recently through a GRACE follow-on. Furthermore, GRACE has restrictions in embedding the local scale GWW due to its coarse resolution and differentiating between shallow and deep groundwater.[82] This is further confirmed from our results obtained in the Bengal basin, where the dominant factor was found to be GWR despite experiencing very high GWW.[14] Notwithstanding the uncertainties and limitations, it can be implicated that there is a trade-off between GWR and GWW in the GWS change in India. Moreover, there has been an unprecedented increase in deep GWW[83] in recent times through deep irrigational wells that produce at least 15 times more groundwater output than shallow irrigation wells in India.[64] The paradigm shift in irrigation practices toward mining deeper groundwater in India occurred due to the persistent decline in GWLs in the past few decades and bypassing contaminated water at shallow levels[84,85] in the region. Thus, due to increased GWW in the deeper part, pumping induced GWR transpires by increasing available GWS[86,87] and return flow.[5,88] Hence, apart from GWW and GWR in the unconfined aquifers, groundwater resource prediction in India is dependent on other factors, such as GWR in deep, confined aquifers, which is a function of the regional flow path and the subsurface structure, and estimating the influence of the cone of depression originating from millions of abstraction wells[65,83] is particularly important at the unconsolidated formations. Therefore, it is challenging to develop a physical model for predicting future groundwater resources, which is essentially needed to frame sustainable water management policies at the microlevel.

3.3. Spatial and Depth-Dependent AI Model Performance. We develop multiple AI-based prediction models (FNN, RNN, LSTM) to simulate and forecast GWLs for future times,

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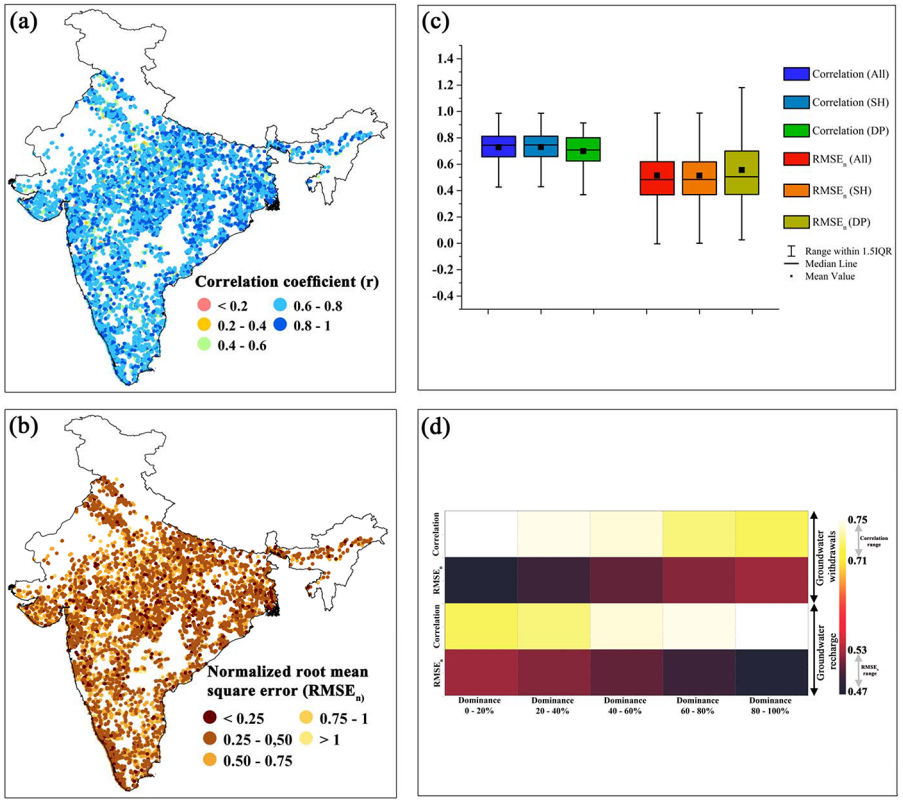


Figure 6. (a) Correlation coefficient and (b) normalized root-mean-square error maps between actual and LSTM predicted groundwater levels of 5367 observation wells; (c) boxplots for the correlation coefficient and normalized root-mean-square error between actual and LSTM predicted groundwater levels for all, SH, and DP observation wells; (d) relationship between LSTM model performance (median values) and dominance of groundwater withdrawals and recharge.

using only GWL as the input. Hence, the relationship between the actual and simulated GWLs (2014−2018) is assessed separately for each of the 5367 observation wells. Although we have developed FNN and traditional RNN based prediction models, the performance of LSTM is found to be the best (Figure 4) in comparison to both RNN (∼59% of $r > 0.6$; ∼61% of $RMSE_n < 0.7$) (Figure 5a,c) and FNN (∼60% of $r > 0.6$; ∼61% of $RMSE_n < 0.7$) (Figure 5b,d). Generally, the results show a strong relationship in terms of the correlation coefficient (r) and the normalized equivalent of the root mean squared error ($RMSE_n$), between the actual and simulated GWLs across the country. However, a marginally weaker relationship is found in northwest India (Indus basin in the west), Bengal basin area, midreaches of the Ganges basin, parts of the Brahmaputra basin, and parts of the Krishna, Cauvery, and Pennar basin of South India (Figures 5 and 6a,b,d). In the GWW dominated regions, the good performance of the LSTM is worth mentioning since both RNN (Figure 5a, c) and FNN (Figure 5b,d) have shown some breakdown in predictability. However, in general, the relationship is marginally weaker in GWW dominated areas (Figure 6d). For LSTM, good correlations ($r > 0.6$, $p < 0.05$) between the actual and the simulated GWLs are observed for ∼86% of the observation wells (Figure S3). Furthermore, ∼84% of the wells show satisfactory $RMSE_n$ of less than 0.7 (Figure S3). The descriptive statistics based on the observation well scale analysis for LSTM is shown in the boxplot in Figure 6c.

We further estimated the depth-dependent well-scale prediction statistics and found that ∼87% of the SH wells and ∼84% of the DP show good correlation ($r > 0.6$, $p < 0.05$). Similarly, ∼81% of the SH wells and ∼80% of the DP wells exhibit satisfactory $RMSE_n$ ($RMSE_n < 0.7$). Thus, the results on the depth-dependent analysis (i.e., SH vs DP observation wells) show that, for both the DP and the SH observation wells, the LSTM performance is good; however, there is a slightly better performance in the case of SH wells. This could be explained by the fact that local scale deep GWW[83] from the irrigation, industrial, or drinking water pumping wells influences the near vicinity observation wells. The deep groundwater abstraction through deep irrigation wells (∼15-fold higher output to shallow irrigation wells)[65] is a common practice to avoid shallow contaminated water and due to continuous dependence of GWL in the region, particularly in the Indus−Ganges−Brahmaputra basin of north India.[90,91] The greater deep groundwater abstraction has led to higher deep GWW in the region. Thus, the observation wells in vicinity of the abstraction wells could be highly affected by the local GWW. In addition to the direct GWW, the irrigation return flow[5] and canal leakage[13] (particularly in the northwest India) could have a significant impact on GWL variation in the study area. Furthermore, GWW impacts the deep confined aquifer at a greater head decline comprising a larger area and a varying equilibrium time, based on the attributes of the confining unit.[92,93] The process could be further complicated by the pumping induced mixing of deep and shallow groundwater.[14] Thus, it could be attributed to the higher deep GWW and model's inability to represent deep/confined aquifer processes. Due to the complex multipart hydrological physical processes, it is problematic to model the hydrologic relationships. Additional knowledge and verifiable data on

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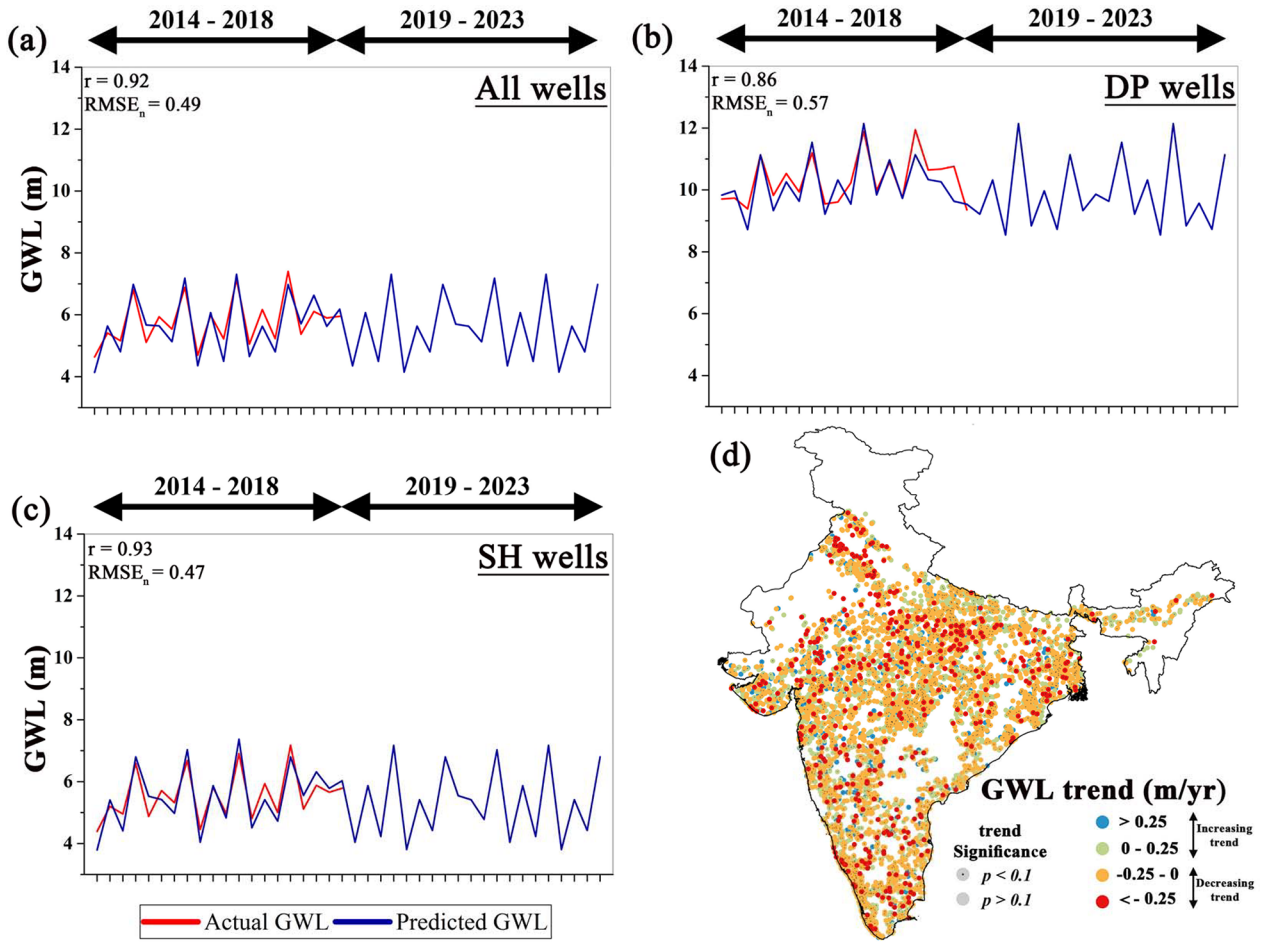


Figure 7. Country-averaged actual and predicted groundwater levels (m bgl) for (a) all, (b) SH, and (c) DP observation wells using LSTM; (d) LSTM based 5-year trends in the in situ groundwater level determined by the Mann–Kendall trend tests for slope significance and Sen's slope estimator for the slope magnitude for 2019 to 2023.

Table 1. Country-Scale Correlation Coefficient (r) and Normalized Root Mean Square Error ($RMSE_n$) between Actual and LSTM Predicted Groundwater Levels for 2014–2018

	correlation coefficient (r)	normalized equivalent of root mean squared error ($RMSE_n$)
all wells	0.92	0.49
SH wells	0.93	0.47
DP wells	0.86	0.57

aquifer lithology and GWW are required to gain a better perception of the subject.

We also computed the actual and simulated (2014–2018) country-averaged GWL time series with future forecasting for the next five years (2019–2024) for All, SH, and DP wells (Figure 7a–c, Table 1). The designed multiscale approach shows similar results from both the well-scale and country-scale prediction statistics, suggesting a good generalization ability. Here, we further identified a statistically significant LSTM based GWL trend ($p > 0.1$) in the forecasting period of 5 years (2019–2023). The forecasted GWL trend (Figure 7d) indicates a higher decline rate in several parts of the country compared to the present long-term decline rates (Figure 1c). The forecasted trend exhibits a large proportion (64%) of observation wells showing a declining trend while the remaining 36% show an increasing trend. In addition to India's observed depleted zone (Figure 1c), our deep learning-based model infers severe GWL decline in regions, including north-central India and south India (Figure 7d).

3.4. Implications for Crop Production and Drinking Water Supply. India's population has a 2-fold increase in the last three decades, with a yearly growth of 2.5% in urban population.[94] In recent times, the higher groundwater dependence of this vast population for drinking water and agriculture purposes leads to unsustainable exploitation in most of the Indian aquifers. Groundwater supports most of the water demand for irrigation and drinking water supply (Figure S4a) in the rural areas[83] (~70% of total population[3]), whereas groundwater is also the predominant source of drinking water in the urban areas[83] in India (Figure S4b), either through municipal supplies or private wells. Groundwater depletion in Punjab, Haryana, Uttar Pradesh, and West Bengal is primarily linked to the higher production of food grains (Figure S5a–c). The total area irrigated in India has been doubled between 1950 and 2015, whereas the area irrigated by wells (which are groundwater-driven) exhibited a 6-fold increase (Figure S5d). In the north-central states of Jharkhand and Madhya Pradesh, the annual yield of food grain production has recorded ~50% and ~60% incremenst, whereas the south Indian states (i.e., Andhra Pradesh, Tamil Nadu, and Kerala) exhibit an ~10%–

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20% increase, which are identified as the probable higher depleting zones in the forecasting models.[95,96] Furthermore, based on the census of 2011, the majority of the drinking water sources are groundwater-dependent (Figure S4c), and there was a significant increase in tap water (52.5%) and hand pumps (20.7%) between 2001 and 2011.[97] Thus, with the diminishing groundwater resources, there is a critical concern for the water managers and administrators at some of the biggest Indian cities that are already struggling to provide adequate water to meet citizens' daily needs.[98] The situation will further exacerbate if it goes uncontrolled. Moreover, India's depleting groundwater resources have a significant implication in crop production and drinking water supply, particularly in the hard-rock peninsular regions with low groundwater storage capacity and drought-prone lands. Groundwater regulations in India are operated through state-level strategies and policies,[99,100] which include bore well registrations, specific depth of bore wells, and protection zone identification.[101] In this context, our forecasted groundwater level output would provide straightforward data to local administrators and policy makers in order to manage groundwater use in the near future.

Here in this study, we assessed different aspects of the human−natural complex system and addressed some of India's key challenges in groundwater resource management. The application and analysis proposed in this study can be employed as an operational tool for efficient and intelligent management and monitoring of groundwater resources, specifically groundwater-fed irrigation dependent regimes in India. However, due to the paucity of data, many of the findings had to be defined qualitatively.[20] Furthermore, knowledge on the extent of spatiotemporal dependencies is limited and further affected by the state of the system. Moreover, AI-based models are built on the premises to improve predictive accuracy. Lack of interpretability has been identified as a potential pitfall of AI-based models. These models are often data-poor and do not integrate the relevant knowledge (e.g., hydraulic properties, topography), which are generally incorporated in physically based models. Hence, the AI-based models have limitations in modeling certain aspects of the physical processes. Therefore, it is necessary to incorporate the information acquired from decades of physical-based modeling into the development of AI-based models. Future works may consider using classical artificial intelligence (AI), guided with physical processes (physically based AI), to gain further process understanding in hydrologic system science problems.[102−104]

5. CONCLUSIONS

Due to the unprecedented decline in GWS in the recent past, a balance between GWR and GWW is critical for India's future groundwater availability. Here, first, we examine GWW and GWR's relative influence in GWS changes and, finally, establish the possible linkage between dominant factors altering the present and future GWLs in India. We conclude that GWW influences GWS in most parts of India, either through direct withdrawals or induced recharge. This is reflected in the rapid decline in India's GWL trends, primarily in northwest India, and parts of the Ganges basin, including the Bengal basin area in eastern India. Furthermore, our deep learning-based LSTM provides good prediction results across India. The performance of LSTM is better than both RNN and FNN. In addition to the present long-term GWL declining trends, our analysis using the most efficient prediction model (i.e., LSTM) also identifies a rapid decrease in GWL trends in parts of north-central India and south India in the 5 years of the forecasting period (2019−2023), which is higher than the estimated present GWL decline rates. Our proposed prediction model incorporating field observations is particularly important since observation wells provide high-resolution ground truth observations than any other major alternatives (i.e., satellite-based observations). We envisage that the output of this study would help the local administrators and policymakers for the sustainable management of groundwater resources.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsestengg.0c00238.

Additional description of the WaterGap model, LSTM, and model evaluation metrics along with additional tables and figures (PDF)

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Notes

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Research Unit (CRU TS v-4.01) at https://crudata.uea.ac.uk/cru/data/hg/. GRACE terrestrial water storage data were obtained from NASA JPL's archive: http://grace.jpl.nasa.gov. The GLDAS data are obtained from GLDAS simulations that are archived and distributed by the Goddard Earth Sciences (GES) Data and Information Services Center (DISC) and can be accessed at https://disc.gsfc.nasa.gov/datasets?keywords=evapotranspiration&sort=-timeRes&page=1&source=Models%2FAnalyses%20OBSERVATION%20BASED. Groundwater withdrawals and groundwater recharge data are taken from the WaterGAP (version 2.2b) simulation, available from Goethe University Frankfurt's archive (https://www.uni-frankfurt.de/45217892/Datens%C3%A4tze___Data_sets).

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